Advances in speech recognition and AI models have enabled automated systems to mediate critical services such as customer support, authentication, and fraud detection. These systems increasingly operate as end-to-end applications on our phones and desktops, where the outputs of speech recognition interfaces are directly consumed by AI agents that interpret intent and take action. While this improves efficiency, it introduces new security and reliability risks: transcription or language understanding errors can propagate downstream, leading to incorrect actions, denied access, frustration or even exploitable behavior. These risks are particularly acute for members of the deaf and hard-of-hearing (DHH) community because their speech often differs in articulation, phonetics, and prosody in ways underrepresented in training data. As a result, DHH users experience elevated transcription and interpretation errors that are not only frustrating, but can trigger incorrect or even unsafe system actions. Worse yet, security mechanisms, such as synthetic speech detection systems, may also misclassify authentic DHH speech as adversarial behavior. This raises a broader question: **how do AI systems behave when benign but underrepresented inputs resemble adversarial ones?** To answer that, we investigate how distributional mismatch in human speech affects the security, robustness, and trustworthiness of audio-driven AI pipelines, and how to mitigate these vulnerabilities.

Intellectual Merit: This project advances a new security paradigm for AI systems by treating distribution shift in human communication as a first-class vulnerability in speech-driven pipelines. Using DHH speech as a representative and high-impact case, we develop a principled framework for understanding and mitigating how benign, underrepresented inputs can trigger unsafe or adversarial-like behavior in modern audio→AI→action systems following four integrated thrusts: (1) **Security analysis of speech-driven pipelines:** model how errors propagate across end-to-end systems and identify conditions under which benign variability leads to unsafe outcomes; (2) **Benchmarking under distribution shift:** construct and release a dataset capturing diverse DHH speech characteristics to evaluate ASR, LLM-based agents, and deepfake detectors under security-relevant metrics; (3) **Quantification of vulnerability surfaces:** develop methods to measure how underrepresented speech induces downstream failures and misclassification in security models, exposing when benign inputs are treated as adversarial; (4) **Mitigation and robust system design:** design defenses spanning data augmentation, representation learning, and fairness-aware objectives, alongside system-level safeguards such as uncertainty-aware decision policies and human-in-the-loop controls.

Broader Impacts: This project addresses foundational challenges at the intersection of security, accessibility, and trust in AI systems**.** As speech interfaces become a primary gateway to essential services, failures in robustness disproportionately exclude DHH individuals from critical infrastructure, including financial systems, healthcare access, and digital public services. This work seeks not only to reduce these barriers, but to establish new principles for **inclusive and trustworthy AI deployment**. By systematically characterizing and mitigating failure modes in modern audio pipelines, the project will generate actionable insights into how AI systems can distinguish between benign variability and adversarial behavior, with a specific focus of reducing misclassification of natural disfluencies as synthetic artifacts and improving speech-driven agents' ability to reliably interpret out-of-distribution articulation, phonetics, and prosody. More broadly, the project **reframes accessibility as a security and robustness problem**, advancing a new perspective in which fairness, reliability, and safety are inherently coupled in AI system design. The project will produce publicly available datasets, benchmarks, and open-source tools that serve as a foundation for the next generation of secure and inclusive speech research. Collaboration with the Georgia Tech Voice+ Lab and DHH community organizations will ensure that these advances are grounded in real-world needs and responsibly developed.